Current System Architecture Analysis

Factual specification of the Genesis Institutional Trading & Strategy Intelligence Platform as implemented in this repository. This document serves as the technical specification for architecture, runtime flows, and diagram models.

Source of truth: `frontend/src`, `backend/app`, `backend/alembic`, `backend/alembic_live`, `backend/pyproject.toml`, `frontend/package.json`, `scripts/`, `trades/`, and `.env.example`.

Analysis & System Specification Date: 2026-10-30.

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1. Current System Components

1. Frontend Client

| Field | Value |

| :--- | :--- |

| Name | Next.js UI |

| Responsibility | Executive Dashboard (index quotes, paper positions, live PnL), Markets (interactive candle charts, non-repainting BOS/CHoCH swings, FVG imbalances, dynamic trendlines, option chain, manual paper orders), Strategies (catalog and 5-Pillar Strategy Analyzer), Strategy Lab (`/strategy-lab` interactive research), Lab (`/lab` ingest, clarify, review crew, simulated trades), Forward-Tested (`/forward-tested` trade journal), Deploy (`/deploy` policy, backtest, approval), Deployments (`/deployments` pause, stop, token-gated kill switch flatten), and Login (`/login`) |

| Technology | Next.js 15, React 19, Tailwind CSS 4, `lightweight-charts` |

| Main files | `frontend/src/app/layout.tsx`, `page.tsx`, `markets/page.tsx`, `strategies/page.tsx`, `strategy-lab/page.tsx`, `lab/page.tsx`, `forward-tested/page.tsx`, `deploy/page.tsx`, `deployments/page.tsx`, `login/page.tsx`, `lib/api.ts`, `lib/market.ts`, `lib/policy.ts` |

| Sync/async | Browser `fetch` over HTTP REST |

| Invoked by | Operator / User web browser |

| Invokes | FastAPI at `NEXT_PUBLIC_API_URL` (default `http://127.0.0.1:8841`) |

| Storage | `localStorage` keys `tsl.actor` and `tsl.token` |

2. Backend API Process

| Field | Value |

| :--- | :--- |

| Name | FastAPI application |

| Responsibility | HTTP REST API for market quotes & candles, chart structure analytics, option chain, paper orders, strategy ingestion & gap clarification, Review Crew executions, live approvals, deployment lifecycles, cryptographic audit chains, notifications, and auth |

| Technology | FastAPI, Uvicorn, Pydantic v2, SQLAlchemy |

| Main files | backend/app/main.py, backend/app/api/routes.py, journal.py, audit.py, live.py, market.py, notifications.py, auth.py, common.py |

| Invoked by | Next.js Frontend and platform automation scripts (GET /api/health) |

| Middleware order | CORSMiddleware → AuditContextMiddleware (X-Request-ID, X-Actor) → RequestLoggingMiddleware (rotating subsystem logger) |

3. Ingest and Extraction Engine

| Field | Value |

| :--- | :--- |

| Name | Strategy Ingest & Extraction |

| Responsibility | Fetch source content from URLs/text and extract executable Markdown strategy documents with rules: blocks |

| Technology | youtube-transcript-api, httpx, BeautifulSoup with lxml, LangChain chat models |

| Main files | backend/app/ingest/youtube.py, instagram.py, doc.py, store.py, backend/app/extract/extractor.py, backend/app/strategies/clarify.py |

| Invoked by | POST /api/ingest/extract, POST /api/ingest/clarify, POST /api/demo/run |

| External deps | YouTube, Instagram, web documents. OpenAI or Anthropic when API keys are configured |

| Storage | ingest_runs, strategies, and strategy_versions in Research DB |

4. Review Crew (LangGraph Research Engine)

| Field | Value |

| :--- | :--- |

| Name | Review Crew |

| Responsibility | Deterministic multi-scenario backtesting, stress testing, capital hard-stop sweeps (2-5%), per-trade forensic reviews, verdict synthesis, and improvement proposals with Indian options cost schedules |

| Technology | LangGraph StateGraph |

| Main files | backend/app/agents/review_crew/graph.py, reviewer.py, hard_stop.py, improver.py, backend/app/engine/sim.py, backend/app/engine/costs.py |

| Invoked by | POST /api/review/run and Strategy Lab |

| Node order | audit → scenarios → simulate → hard_stop → trade_review → verdict → improve |

| Storage | reports, simulation_runs, sim_trades, trade_reviews, strategy_improvements |

5. The 5 Autonomous Trading Crews (backend/app/trading/)

The live trading architecture is decoupled into 5 specialized autonomous crews managed by scripts/live_scanner.py and scripts/genesis.sh:

1. **PreMarketCrew (backend/app/trading/pre_market_crew/)**:

- PreMarketManager: Runs daily from 08:30 to 09:14 IST.

- `GapAgent`: Assesses GIFT NIFTY vs Previous Day Close (PDC) gap bias and opening trap odds.

- `GammaAgent`: Analyzes option chain Call/Put walls, PCR, and dealer pinning Max Pain.

- `EventRiskAgent`: Checks macro economic calendar (RBI MPC, CPI, FED) for event blackout restrictions.

- Outputs `trades/pre_market_plan.json` and dispatches the 09:05 IST Battleplan to Telegram.

2. **ScannerCrew (`backend/app/trading/scanner_crew/`)**:

- `ScannerManager`: Runs continuously during market hours (09:15 - 15:30 IST).

- `FeedAgent`: Delta-only multi-timeframe polling (1m, 2m, 5m, 15m, 30m) with ETF session VWAP proxy.

- `LevelsAgent`: Pins dynamic PDH, PDL, PDC, Opening Range (OR-30: 09:15-09:45), and psychological strike levels.

- `ConsolidationAgent`: Midday compression box & ATR deadband detector (Lunch Lull 11:30-13:00 filter).

- `NewsAgent`: Evaluates live news sentiment and macro catalyst risks.

- `DetectorAgent`: Multi-strategy pattern scanner for Breakout & Retest, Liquidity Sweep (PDH/PDL raid), Trendline Reversal, Dual-Trendline Squeeze Detection, and Dynamic Lookout Horizons.

- Emits real-time `ScanSnapshot` and verified `CandidateSetup` objects.

3. **TradeCrew (`backend/app/trading/crew/`)**:

- `TradeManager`: Master live trade orchestrator.

- `TrendAgent`: Calculates macro trend slope and aligns dynamic R:R targets (+1.8R to +4.5R).

- `OrderFlowAgent`: Detects institutional whale volume spikes and absorption traps to issue pre-entry vetoes.

- `SniperAgent`: 1M micro-wick execution trigger, shrinking stops to 15–25 pts for asymmetric 1:5R+ expansion.

- `RiskAgent`: Enforces 3-Tier sizing, daily loss limits, consecutive loss circuit breakers, and 50% Capital Safety Wall.

- `EntryAgent`: Executes Directional Thesis formulation with strictly 2-OTM strikes ($\Delta \ge 0.35$ floor).

- `Idempotent Guard`: SHA-256 order signature authorization key preventing duplicate fills and double orders.

- `ExitAgent`: Unconditional Target 3 full close, 5M structural trailing stop, 30-min Charm stagnation time-stop, and opposing sweep auto-flip.

- `AlertAgent`: Dispatches live Telegram trade tickets with complete Directional Thesis and threaded CMP heartbeat updates.

- `JournalAgent`: Manages active trade cache and commits atomic ACID records to `trades.db` SQLite SSOT.

4. **ExecutionCrew (`backend/app/trading/execution_crew/`)**:

- `ExecutionManager`: Coordinates order routing and broker bridges.

- `SmartRoutingAgent`: Tranche order execution (30% market fill + 70% limit chase) to minimize slippage.

- `BrokerBridgeAgent`: Pluggable adapter bridge for Paper and Live broker execution.

- `ReconciliationAgent`: Zero-drift position reconciliation ensuring database state matches broker holdings.

5. **PostMarketCrew (`backend/app/trading/post_market_crew/`)**:

- `PostMarketManager`: Dispatched automatically at 15:31 IST daily.

- `AutopsyAgent`: Reconstructs the day's intraday tape, audits exit efficiency, and logs missed setups.

- `ParameterAgent`: Executes walk-forward edge testing and parameter tuning for tomorrow's session.

- `SafetyWallAgent`: Sweeps net realized profits and locks 50% permanently into the protected Capital Floor.

- `CodeQualityAgent`: Audits application logs, system warnings, and latency bottlenecks.

- `BriefingAgent`: Writes `reports/daily_audit_YYYY-MM-DD.md` and dispatches the Executive Digest to Telegram.

6. Architecture C Protective Engines (`backend/app/trading/crew/`)

Architecture C ("The Optimized Hybrid Engine") adds 6 specialized protective engines directly into the execution path:

1. **Directional Thesis Engine**: Generates `primary_bet`, `targeted_pool`, `invalidation_price`, and `thesis_conviction`.

2. **Delta Floor Engine**: Guarantees estimated option $\Delta \ge 0.35$ on strictly 2-OTM strikes, preventing Charm decay.

3. **Gamma Wall Runway Gate**: Checks ≥ 80 pts open runway to opposing dealer gamma walls and vetoes <20 pt vacuum traps.

4. **Charm & Stagnation Time-Stop**: Exits trades stagnating for 6 five-minute bars (30 min) without reaching Target 1.

5. **Idempotent Order Guard**: Computes `SHA-256(date_strategy_dir_strike_price)` to suppress duplicate orders across live and paper.

6. **Session & VIX Gatekeeper**: Filters trading to India VIX 11.5–22.0, blocks 13:00–13:30 Gate E lunch lull, and enforces 15:10 IST square-off.

*(Total System Components: 29 Modular Crew Agents/Managers + 6 Architecture C Engines + 15 LangGraph State Nodes + 2 Strategic Diagnostics + 1 Supervisor Daemon = **53 Functional Components**)*

6. TradeStore SSOT & Excel Synchronization (`backend/app/db/trade_store.py`)

- High-performance SQLite database (`data/trades.db`) operating in WAL (Write-Ahead Logging) mode.
- Provides atomic transactions for concurrent multi-process access (Scanner, FastAPI, Exporter).
- `scripts/maintain_trades_excel.py` automatically synchronizes the SQLite store with institutional Excel workbooks in `trades/`:

- `trades/live_and_forward_trades.xlsx`: Master executive dashboard and live trade ledger.

- `trades/todays_live_trades.xlsx`: Isolated intraday session trade ledger.

- `trades/by_strategy/*.xlsx`: Strategy-specific workbooks.

7. Non-Repainting Market Structure Engine (`backend/app/market/chart_analytics.py`)

- Computes non-repainting pivot highs and lows using rolling 2-left / 2-right bar verification.
- Identifies Market Structure Breaks (`BOS`) and Changes of Character (`CHoCH`).
- Maps Equal Highs (`EQH`), Equal Lows (`EQL`), and Fair Value Gaps (`FVG`).
- Calculates dynamic ascending support and descending resistance trendlines.

8. Platform Management CLI (`scripts/genesis.sh`)

Unified bash control script managing background services:

- `./scripts/genesis.sh start`: Starts Backend (8841), Frontend (3000), and Live Scanner Daemon.
- `./scripts/genesis.sh stop`: Gracefully terminates all background services.
- `./scripts/genesis.sh restart`: Clean restart of all services.
- `./scripts/genesis.sh status`: Detailed PID health and HTTP status check.

- `./scripts/genesis.sh logs [backend|frontend|scanner]`: Tails process logs in real-time.

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2. Multi-Tier Persistence Architecture

```
flowchart TD
    subgraph StorageTier["Genesis Multi-Tier Persistence Layout"]
        direction TB
        R_DB[("1. Research DB (DATABASE_URL)<br/>SQLite/Postgres · Alembic 0006<br/>strategies, versions, reports, sim_trades,<br/>trade_reviews, paper_orders, notifications,<br/>SHA256 audit_events")]
        L_DB[("2. Live DB (LIVE_DATABASE_URL)<br/>SQLite/Postgres · Alembic live 0005<br/>deployed_strategies (frozen hashes), deployments,<br/>orders, fills, live_trades, credentials (Fernet),<br/>SHA256 live_audit_events")]
        T_DB[("3. TradeStore SSOT (data/trades.db)<br/>SQLite WAL Mode<br/>live_and_forward_trades, PnL,<br/>active_strategy_params, safety_wall_state,<br/>daily_risk_config")]
        M_DB[("4. Market DB (MARKET_DATABASE_URL)<br/>SQLite · Unaudited Reference Store<br/>index_bars, option_eod, events, news")]
        XL[("5. trades/ Excel Workbooks<br/>live_and_forward_trades.xlsx<br/>todays_live_trades.xlsx<br/>by_strategy/*.xlsx")]
    end
```

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3. Present vs Absent Implementation Status

| Feature / Capability | Status | Implementation Evidence |

| :--- | :--- | :--- |

| **5 Autonomous Trading Crews** | Present | `backend/app/trading/{pre_market_crew, scanner_crew, crew, execution_crew, post_market_crew}/` |

| **Multi-Timeframe Live Scanner** | Present | `scripts/live_scanner.py`, `scripts/genesis.sh` |

| **TradeStore SQLite SSOT (WAL)** | Present | `backend/app/db/trade_store.py`, `data/trades.db` |

| **Excel Synchronizer** | Present | `scripts/maintain_trades_excel.py`, `trades/` |

| **3-Tier Sizing & 50% Safety Wall** | Present | `backend/app/trading/scaling_law.py`, `trades/daily_risk_config.json` |

| **Market Structure Engine (BOS/FVG)** | Present | `backend/app/market/chart_analytics.py`, `frontend/src/components/CandleChart.tsx` |

| **Strategy Lab & Ingest Engine** | Present | `backend/app/ingest/`, `backend/app/extract/`, `frontend/src/app/lab/` |

| **Deterministic Review Crew** | Present | `backend/app/agents/review_crew/graph.py` |

| **Telegram & Email Dispatch** | Present | `backend/app/alerts.py`, `backend/app/trading/crew/alert_agent.py` |

| **Paper & Sandbox Brokers** | Present | `backend/app/brokers/paper.py`, `backend/app/brokers/sandbox.py` |

| **Dual Cryptographic Hash Chains** | Present | `backend/app/audit/`, `backend/app/live/audit.py` |

| **LLM Trade Execution Authority** | **Absent (By Design)** | Order paths are strictly deterministic in Python code; LLMs have zero trade initiation capability |

| **Exchange Broker Adapters** | Open Extension | Plug into `BrokerBridgeAgent` / `backend/app/brokers/` |

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4. Visual Diagrams Reference

- System Architecture
- 5 Autonomous Crews Ecosystem
- Live Scanner & Execution Flow
- Review Crew & Trade Crew Workflow
- Database Schema Layout